

MODULE-1: Robotic Process Automation and Record & Play



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Certificate

This is to certify that the e-book titled “Robotic Process Automation and Record & Play” comprises all elementary learning tools for a better understating of the relevant concepts. This e-book is comprehensively compiled as per the predefined eight parameters and guidelines.



Signature

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Unit I

- **Contents**

Robotic Process Automation: Scope and techniques of automation, About UiPath

Record and Play: UiPath stack, Downloading and installing UiPath Studio, Learning UiPath Studio, Task recorder, Step-by-step examples using the recorder

Recommended Books

- **Text Books And Reference Books**

Unit I	Prerequisites	Linkage
	Sem. I	Sem VI
Learning Robotic Process Automation	Programming Logic	Project

Unit I

- **Robotic Process Automation:** Scope and techniques of automation, About UiPath
- **Record and Play:** UiPath stack, Downloading and installing UiPath Studio, Learning UiPath Studio, Task recorder, Step-by-step examples using the recorder.

Q.1 What is automation?

Ans:

1. Automation refers to the use of technology and machinery to perform tasks or reduced human intercession.
2. Automation in simple words is technology that deals with the application of machines and computers to the production of goods and services. This helps in getting work done with little or no human assistance.
3. There are various techniques used and available to automate steps and processes in an organization where software systems are being used to accomplish certain tasks.
4. Developing an inventory management software system is called software development, while programming a step so that no more human intervention is required is called automation.

Q.2 What should be automated?

Ans:

1. There are a few aspects that must be taken into consideration for choosing automation candidates.
2. The following processes should be automated:
 - Repetitive steps
 - Time-consuming steps
 - High-risk tasks
 - Tasks with a low-quality yield
 - Tasks involving multiple people and multiple steps.
 - And everything else!

Q.3 What can be automated?

Ans:

1. To automate something, it needs to have the following characteristics:
 - Well defined and rule-based steps
 - Logical
 - Input to the task can be diverted to the software system.
 - Input can be deciphered by software systems with available techniques.
 - The output system is accessible.
 - Benefits are more than the cost.
2. If something has this characteristic it can be automated.

Q.4 What are the techniques of automation?

Ans:

1. **Custom software:** Developing new software to perform repetitive tasks.
2. **Runbook:** Runbooks are typically used for IT-based operations. They are a compilation of a set of commands or tasks that are performed for maintenance and other types of activities. Runbooks can be offline as well, often referred to as run commands for performing sets of tasks.
3. **Batch:** Batch files used to be very popular. They used to compile a sequence of commands that could be run by a single click or command. They can also be scheduled to be run at a specific time using the scheduler.
4. **Wrapper:** Wraps around existing software or hosts client applications. The wrapper monitors activities in a client app and performs actions based on rules.
5. **Browser automation:** Grease monkey and many other web macro software helped in browser-based automation. It can be used to read from a website and save to a database. It can also write to fields based on rules. Using this technique, a whole website can be changed, and components can be added or removed from the website. Sometimes, it is also referred to as web scripting or web injections.
6. **Desktop automation:** Traditionally, desktop automation used to mean that multiple screens on a desktop were woven together to present a single screen, and if there was some data transfer from one screen to another, it could be done automatically. Recently, assisted Robotics process automation has also been considered for desktop automation by some companies. Database/web service.
7. **Integration:** In database integration, we read/write to a client database directly.

Q.5 What is Robotic Process Automation?

Ans:

1. Robotic process automation (RPA) is a transformational burgeoning area. Robots in Robotic process automation means software programs that mimic human actions.
2. In simple words, RPA involves the use of software that mimics human actions while interacting with applications on a computer and accomplishing rule-based tasks. This often requires reading from and typing, or clicking on existing applications that are used to perform the given tasks.
3. Some of the technologies being adopted with RPA are as follows:
 - Machine learning
 - Natural language processing

- Natural language generation
- Computer vision

4. With the inclusion of the preceding technologies, sometimes it is also referred to as intelligent automation.

Q.6 What can RPA do?

Ans:

- RPA has matured beyond doing mundane repetitive tasks and is seen as a transformational technology that can bring tremendous value to the organization adopting it.
- The ability to create full audit trails is significant for improving the quality of work being done and eliminating human error.
- Once trained, these Robots will perform tasks with the same precision repeatedly. These Robots can interact with applications irrespective of the technologies on which the applications are built.
- They can work with popular ERP applications such as SAP, Oracle, or Microsoft Dynamics, and BPMs such as Pega systems and Appian.

Q.7 What are the industries they can benefit from RPA?

Ans:

- Business process outsourcing (BPO): With RPA and its benefits of reduced costs, the BPO sector can now depend less on outsourced labor.
- Insurance: The complexity and number of tasks that must be managed in the insurance sector, from managing policies, to filing and processing claims across multiple platforms, provides an ideal environment for the use of RPA technology.
- Financial sector: From day-to-day activities and handling an enormous amount of data, to performing complex workflows, RPA has been helping to transform this sector into an efficient and reliable one.
- Utility companies: These companies (such as gas, electricity, and water) deal with a lot of monetary transactions and can leverage RPA to automate tasks such as meter reading, billing, and processing customer payments.
- Healthcare: Data entry, patient scheduling, and more importantly billing and claims processing, are important areas where RPA can be used. RPA will help in optimizing patient appointments, sending them automatic reminders of their appointments, and eliminating human error in patient records. This

leaves workers to focus more on the needs of the patients and leads to improved patient experience.

Benefits of Using RPA: https://www.youtube.com/watch?v=uGq8gCoOk_s

Q.8 What are the benefits of RPA?

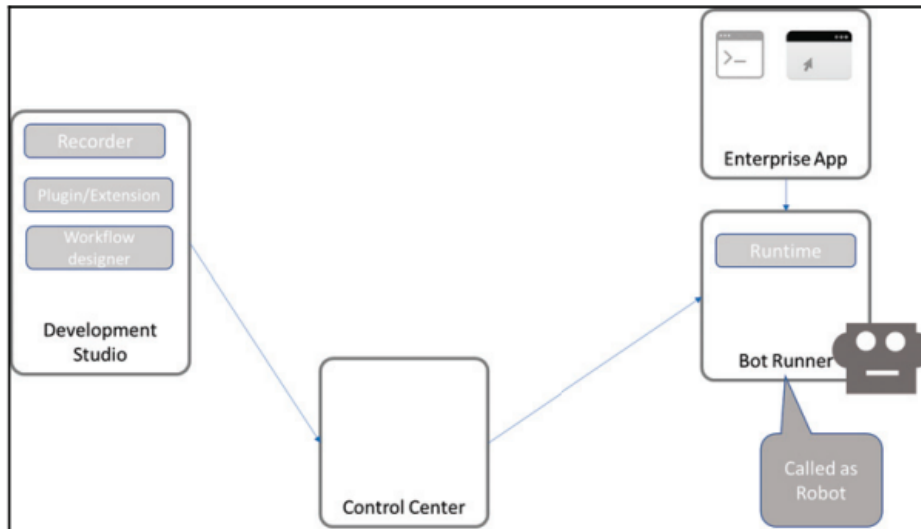
- **Higher quality services, greater accuracy:** Every step in the automation process is recorded, making it faster to pinpoint errors with ease. The reduction, or removal of, errors also mean greater accuracy of data, leading to better quality analytics and hence better decision making.
- **Improved analytics:** Using analytics on the collected data such as transaction received time, transaction complete time, and predictions can be made for the incoming volume and ability to complete the tasks on time.
- **Reduced costs:** Increased availability and productivity means the cost of operations is reduced tremendously. The speed of the work being performed coupled with multitasking results in further reductions in cost.
- **Increased speed:** Robots are very fast and sometimes the speed of execution must be reduced to match the speed and latency of the application on which these Robots work.
- **Greater compliance:** A full audit trail is one of the highlights of RPA and can result in greater compliance. These Robots do not deviate from the defined set

of steps to be taken while doing a task and hence it will certainly result in better compliance.

- **Agility:** More Robots can be deployed to perform the same task easily. Redeployment of resources does not require any kind of coding or reconfiguration.
- **Comprehensive insights:** Robots can tag transactions to use them later, in reports for business insight. By using these business insights, better decisions can be made for the improvement of the business. This recorded data can also be used for forecasting.
- **Versatility:** RPA is applicable across industries performing a wide range of tasks - from small to large businesses, simple to complex processes.
- **Simplicity:** Since automation is carried out by people from within the department or area of work, no requirements are lost in translation between the business unit and the development team, which may have happened otherwise in traditional automation.
- **Scalability:** Whether one requires an increase or reduction in the virtual workforce, Robots can be quickly deployed at zero or minimum costs while maintaining consistency in the quality of work.
- **Time savings:** A technological upgrade makes it much easier and faster for the virtual workforce to adapt to the changes. This can be done by bringing about modifications in the programming or introducing new processes. For humans, it is difficult for them to learn and get trained in something news-breaking from the old habit of performing repetitive tasks.
- **Non- invasive:** RPA helps in reducing risks and complexities that would arise in the case of traditional IT deployments.
- **Better management:** RPA allows for managing, deploying, and monitoring Robots through a centralized platform. This also lessens the need for governance.
- **Better customer service:** Since Robots can work around the clock, capacity increases. This leaves humans to focus on customer service and satisfaction. Also, a better quality of services delivered to customers at faster speeds greatly boosts customer satisfaction.
- **Increased employee satisfaction:** With repetitive, dreary tasks now being taken over by the virtual workforce, employees are not just relieved of their workload, but can also engage in better quality work that requires the use of human capabilities and strengths such as emotional intelligence, reasoning, or tending to customers.

Q.9 What are the components of RPA?

Ans:



- The following are the basic or core components of RPA:
- **Recorder:** The recorder is the part of the development studio that developers use to configure the Robots. It is like the macro recorder in Excel, the bot recorder in any platform, records steps. It records mouse and keyboard movements on the UI and this recording can be replayed to do the same steps again and again. This enables rapid automation. This component has played a very big role in the popularity of RPA.
- **Development Studio:** The development studio is used by developers to create Robot configurations or train the Robots. Using the development studio, a set of instructions and decision-making logic is coded for Robots to execute. Some platforms provide flow-charting capabilities such as Visio, so it becomes very easy to plot steps in a process, whereas some other platforms require coding. In most studios, to do commercial development, developers need to have a fair amount of knowledge of programming, for example, loops, if else, variable assignment, and so on.
- **Plugin/Extension:** Most platforms offer many plugins and extensions to ease the development and running of bots. In many applications, such as Java SAP, it is not easy to individually identify controls of the UI through traditional techniques. RPA vendors have developed plugins and extensions to help with these issues.
- **Bot Runner:** This is also referred to as the Robot, other components make it run.
- **Control Center:** The objective of the control room is to provide Robot management capabilities. It monitors and controls a robot's operation in a

network. It can be used to start/stop Robots, make schedules for them, maintain, and publish code, redeploy Robots to different tasks, and manage licenses and credentials.

Q.10 What are the different types of RPA platforms?

Ans:

- Automation Anywhere: Automation Anywhere helps to automate business processes for companies. They focus on RPA, cognitive data (machine learning and natural language processing), and business analytics.
- UiPath: UiPath is an RPA technology vendor who designs and delivers software that helps automate businesses.
- Blue Prism: Blue Prism aims to provide automation that enterprises can use according to their needs. Blue Prism aims to do this by providing automation that is scalable, configurable, and centrally managed.
- WorkFusion: WorkFusion offers automation that is based on RPA and machine learning. It delivers software as a solution for automating high volume data.
- Thoughtonomy: Thoughtonomy delivers software that helps automate business and IT processes.
- KOFAX: Kofax's Kapow RPA platform is capable of automating and delivering processes that are repetitive and rule-based. It uses Robots for extracting and consolidating information. The software platform consists of a management console to deploy and manage bots, Robot performance, and a monitoring system. This software can also group together high priority tasks that should be completed first by the Robot during times of high workload.

Q.11 Write a short note on UiPath?

Headquartered in Bucharest, UiPath is an RPA vendor that provides software to help organizations automate their business processes.

- a. The company aims to do away with repetitive and tedious tasks, allowing humans to engage in more creative and inspiring activities.
- b. UiPath was founded by Daniel Dines, who is the CEO. It has offices in London, Bucharest, Tokyo, Paris, Singapore, Melbourne, Hong Kong, and Bengaluru.
- c. With clients spread across the world, from North America to the United Kingdom, Continental Europe to Asia Pacific countries, the company has shown remarkable growth in the last year, both in terms of revenue and its workforce.

- d. Today, its software is being widely used to automate business processes. However, the IT sector is also gradually embracing UiPath's software. Major clients of UiPath in the industry include BFSI, Telecom and media, healthcare, retail and consumer, and manufacturing. With UiPath automation software, one can configure software Robots to mimic human action on the user interface of computer systems. The basic components of the UiPath RPA platform are in line with what was explained in Components of RPA, these components are necessary for enterprise deployment. The components of the UiPath platform are UiPath Studio, UiPath Robot, and UiPath Orchestrator, see the following sections.

Q.12 What are the components of UiPath? OR Explain UiPath Stack.

There are three basic components in UiPath:

1. UiPath Studio
2. UiPath Robot
3. UiPath Orchestrator

UiPath Studio: UiPath Studio is the development environment of UiPath. It is the primary tool to develop UiPath Robots. It can be used to configure steps of a task or launch a full recorder to record a sequence of steps. The recording facility in the Studio is a game-changing feature for RPA tools. Its simplicity lets even nontechnical business users design/record steps of a process.

a) This studio lets the user configure Robots, that is, develop steps to perform tasks visually.

b) Most of the configuration and coding in UiPath is visual. By using the drag-drop facility from the toolbox, you may write a whole sequence of workflows to perform a set of tasks by Robots.

c) These steps look like a data flow diagram and are very easy to understand. It is one of the simplest visual flow diagramming tools. Most of the time, in an enterprise environment you will receive process maps to understand the flow of work, which you will use to develop Robots. The studio gives the same look and feel as a workflow.

2. UiPath Robot:

UiPath Robot is a Windows service that can open interactive/non-interactive window sessions to execute processes or a set of steps, developed or recorded using UiPath Studio. Sometimes, it is also called an execution agent as it executes automation projects, or a runtime agent as it executes instructions generated by developing or recording processes in UiPath Studio. The most acceptable

nomenclature is Robot. These Robots can be controlled by Orchestrator, which is part of the Enterprise Edition. There is an option at installation to de-link these Robots from Orchestrator and work independently on the desktop. In most of our examples, we will refer to the Community Edition, which does not have Orchestrator, and the installed Robot will work independently in user mode. When installed in user mode, these Robots have the exact same rights as the user. If you opted for Orchestrator, you could control Robots irrespective of whether it is installed on a user machine, in user mode, or on a server.

The following are types of Robots:

- i. **Attended:** It operates on the same workstation as a human to help the user accomplish daily tasks. It is usually triggered by user events. You cannot start a process from Orchestrator on these types of Robots, and they cannot run under a locked screen.
- ii. **Unattended:** It can run unattended in virtual environments and can automate any number of processes. In addition to the Attended Robot's capabilities, this Robot is responsible for remote execution, monitoring, scheduling, and providing support for work queues.
- iii. **Free:** It is similar to Unattended Robots but can be used only for development and testing purposes, not in a production environment.

3. UiPath Orchestrator:

UiPath Orchestrator is a server-based application that lets you orchestrate your Robots, hence the name Orchestrator.

Orchestrator's main capabilities:

- * It helps in creating and maintaining the connection between Robots
- *It ensures the correct delivery of the packages to Robots
- *It helps in managing the queues
- *It helps in keeping track of the Robot identification
- *It stores and indexes the logs to SQL or Elasticsearch

Behind the scenes, Orchestrator Server uses:

- *IIS Server
- *SQL Server
- *Elasticsearch Kibana

We have the option to store information or logs in an SQL database or Elasticsearch (which is based on Lucene). If the size of the data is small, then SQL will be preferred. However, if we have a large chunk of data and we also need some

analysis on that data, it becomes tedious. Here, text search engine tools such as Lucene come into play.

Q.13 What are the different variations available for RPA?

Ans) The UiPath platform is available in two variations:

- i. **Enterprise Edition:** This edition is suitable for large companies starting their RPA projects and looking to scale their robot deployments in the future. It is integrated with UiPath Orchestrator.
This version can be updated by visiting the UiPath website and by downloading the newest version of the UiPath platform installer. Running the installer automatically replaces all the old files without modifying any of your settings.
- ii. **Community Edition:** This is suitable for individual developers and small organizations with fewer employees. The Community Edition is always up-to-date, and it automatically updates itself as soon as a new version is available.

Q.14 What are the different types of Robots?

The following are types of Robots:

- **Attended:** It operates on the same workstation as a human to help the user accomplish daily tasks. It is usually triggered by user events. You cannot start a process from Orchestrator on these types of Robots, and they cannot run under a locked screen.
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Q.15 What are the main capabilities of Orchestrator?

Orchestrator's main capabilities:

- It helps in creating and maintaining the connection between Robots.
- It ensures the correct delivery of the packages to Robots.
- It helps in managing the queues.
- It helps in keeping track of the Robot identification.
- It stores and indexes the logs to SQL or Elasticsearch

Q.16 What are the main types of projects supported by UiPath Studio?

Ans) The main types of projects supported by UiPath Studio are as follows:

- **Sequence:** This is suitable for simple actions or tasks. It enables you to go from one activity to another, without interfering with your project. It consists of various activities. Creating sequences is also useful for debugging purposes. One activity from a particular sequence can easily be tracked. The Basic type of project can be started using the Blank option in the start tab and then adding the sequence in the diagram from the toolbox.
- **Flowchart:** This is suitable for dealing with more complex projects. It enables you to integrate decisions and connect activities. To start this kind of project, choose the Flowchart - Simple Process option from the new project menu.
- **Assistant:** This is suitable for developing attended or Front Office Robots: sometimes these Robots are called assistants. To start this kind of project, choose the Assistant - Agent Process Improvement option from the new project menu.
- **State machine:** This is suitable for very large projects that use a finite number of states in their execution, triggered by a condition. To start this kind of project, choose the Process - Transaction Business Process option from the new project menu.

New



Blank

An empty project. A diagram is a graphical representation of a business process.



Simple Process

Model a process as a flowchart diagram.



Agent Process Improvement

Trigger an automation in response to a mouse or keyboard user event.



Transactional Business Process

Model a business process as a state machine diagram.

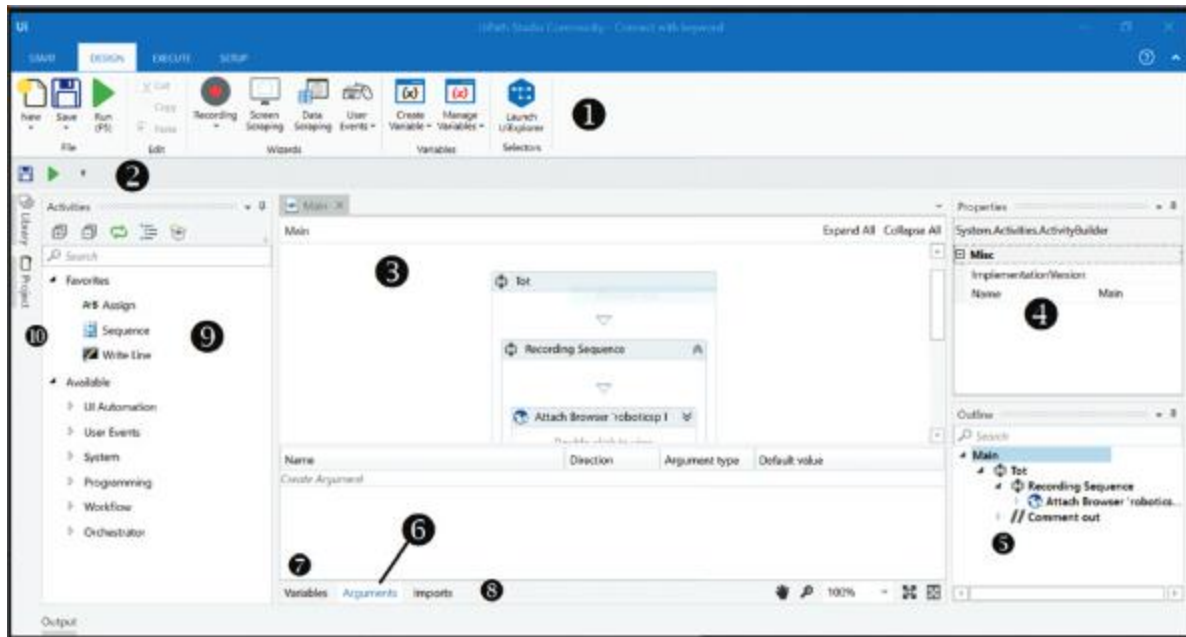
UiPath “Together”: Automation in a Fast-Changing World:
<https://www.youtube.com/watch?v=iWgH5lhVXNk>

Q.17 What is the user interface in UiPath Studio? What are the different panels available in UiPath studio? Explain any five.

Ans) The user interface When you first open UiPath Studio, you are directed to the page shown in the following screenshot:



You can either open an old project or create a new one. Let us say we are making a new project. We click on Blank and name it. We will then be directed to a screen, which will display the following:

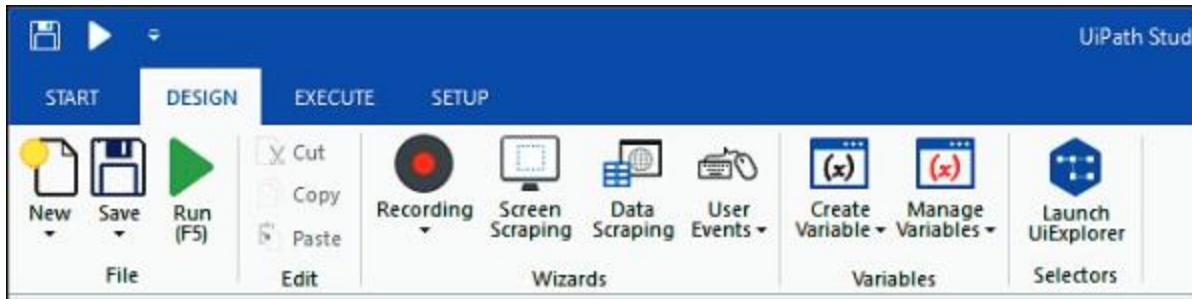


- 1.The Ribbon
- 2.Quick Access Toolbar
- 3.Designer panel
- 4.Properties panel
- 5.Outline panel
- 6.Arguments panel
- 7.Variable panel
- 8.Import panel
- 9.Activity panel
- 10.Library panel
- 11.Project panel
- 12.Output panel

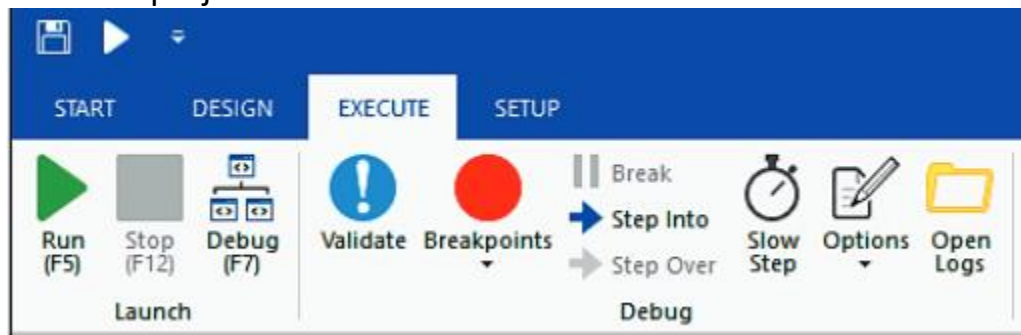
The Ribbon

This panel located at the top of the user interface and consists of four tabs:

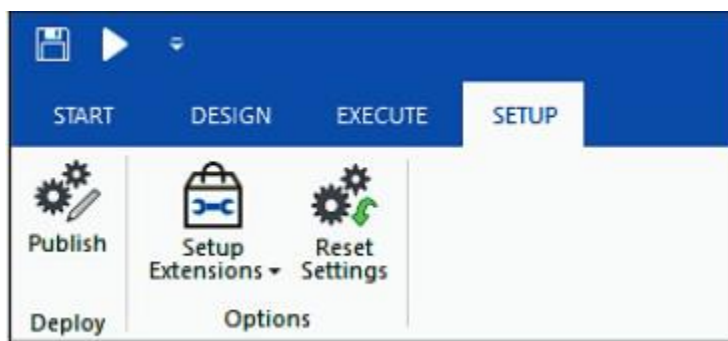
- **START:** This is used to start new projects or to open projects previously made.
- 1. **DESIGN:** This is to create new sequences, Flowcharts, or state machines, or to
- 2. **manage variables:**



3. EXECUTE: This is used to run projects or to stop them, and also to debug projects:



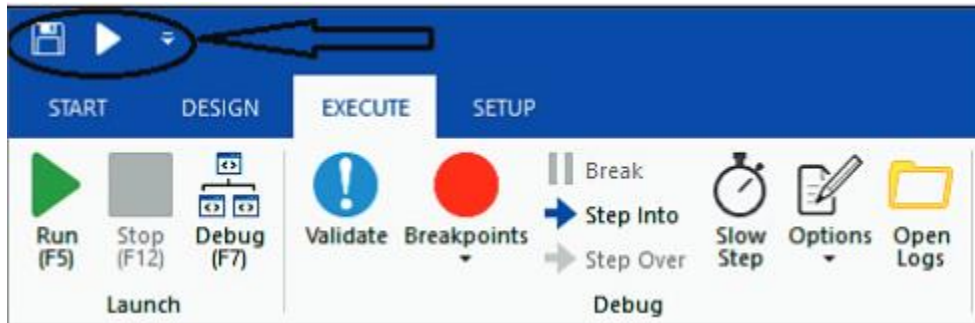
4. SETUP: This panel is for deployment and configuration options; it has three tools available:
 1. Publish: This is used to publish a project or create a shortcut for it and schedule tasks
 2. Setup Extensions: This is used to install extensions for Chrome, Firefox, Java, and Silverlight Reset
 3. Settings: This is used to reset all settings to defaults:



The Quick Access Toolbar

This panel gives the user a shortcut to the most used commands. One can also add new commands to this panel. This is located above the Ribbon on the user interface.

The Quick Access Toolbar has been circled in the following screenshot and is indicated by the arrow:



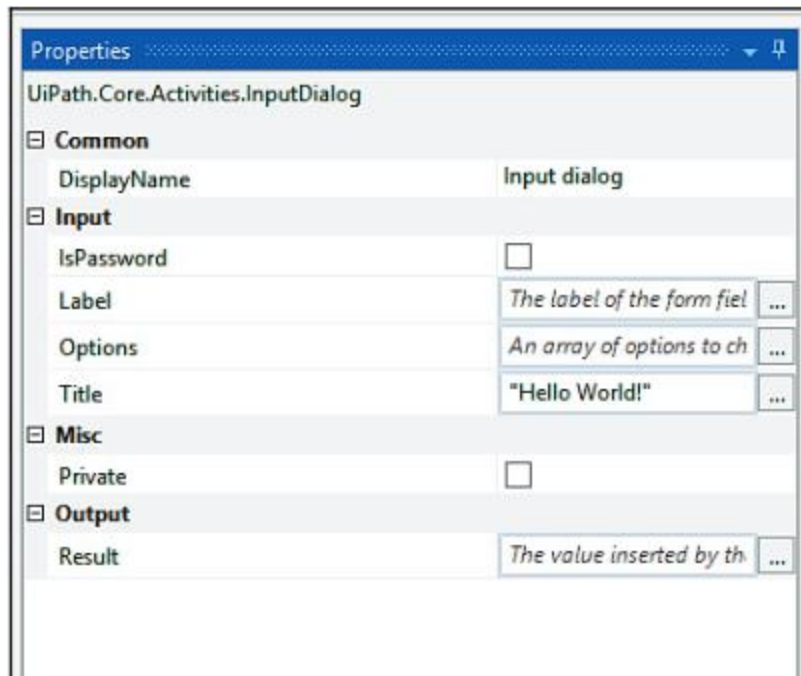
It can be moved above or below the Ribbon. By default, there are two buttons available, Save and Run, which are also available in the DESIGN tab of the Ribbon.

Designer panel

This is the panel where one defines the steps and activities of the projects. It is where a developer does most of the things to record activities or manually drop activities on the canvas. In UiPath, this is equivalent to the code windows of Microsoft Visual Studio. When we develop a robot, this is the window where we will be organizing various activities in a flow or chain to accomplish a task. The project a user makes is clearly displayed on the Designer panel and the user has the option of making any changes to it.

Properties panel

The panel located on the right-hand side of the user interface is for viewing the properties of the activities and for making any changes, if required. You need to select an activity first and then go to the Properties panel to view or change any of its properties:

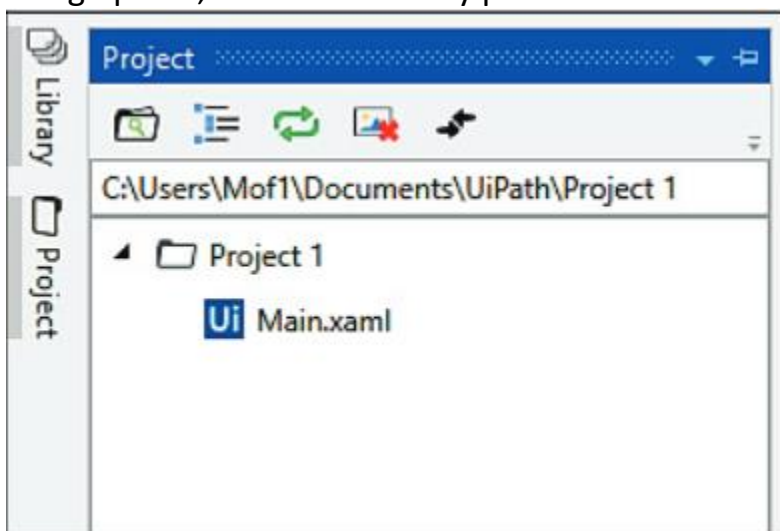


Activities panel

Located on the left-hand side of the user interface, this panel contains all the activities that can be used in building the project. The activities can easily be used in making a project by simply dragging and dropping the required activity into the required location in the Designer panel.

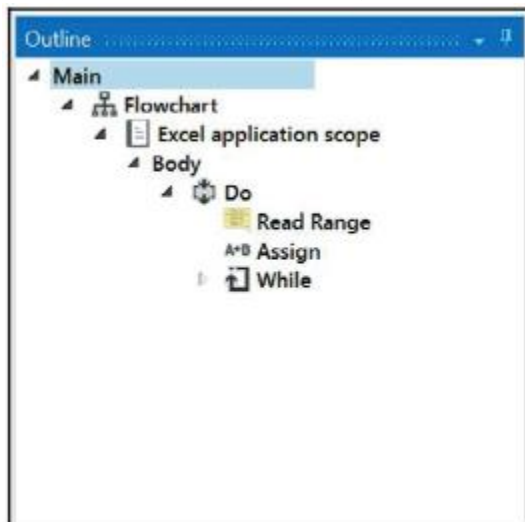
Project panel

With the Project panel, you can view the details of your current project and open it in a Windows Explorer window. It is located on the extreme left-hand side of the design panel, below the library panel:



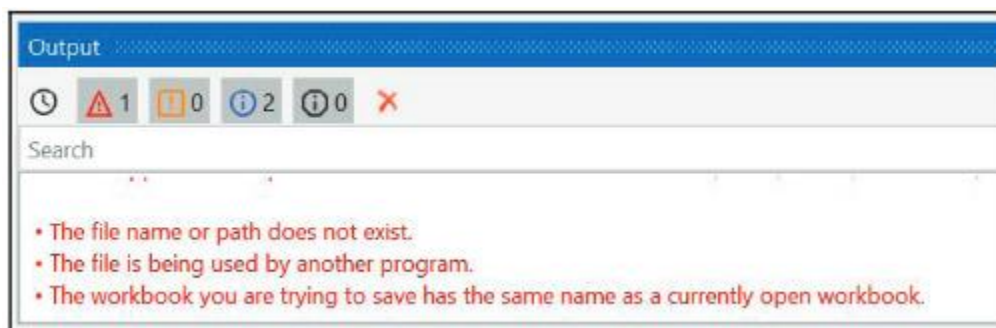
Outline panel

As the name suggests, this panel gives a basic outline of the project. The activities that makeup the workflow is visible in this panel. Using this, you may see a high-level outline of the project and you can drill down to see deeper. This panel is especially helpful of large automation projects, where one may otherwise have a tough time going through it:



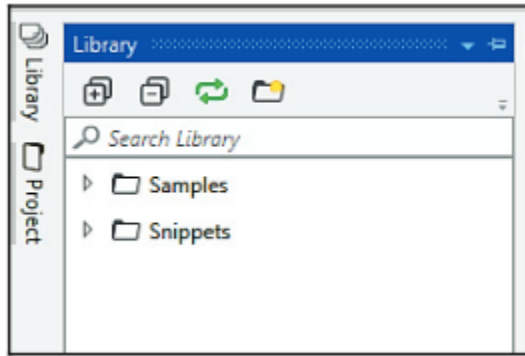
Output panel

This panel displays the output of the log message or write line activities. It also displays the output during the debugging process. This panel also shows errors, warnings, information, and traces of the executed project. It is very helpful during debugging. The desired level of detail can be changed in Execute | Options | Log activities:



Library panel

With this panel, you can reuse automation snippets. It is located on the extreme left-hand side of the Designer panel:



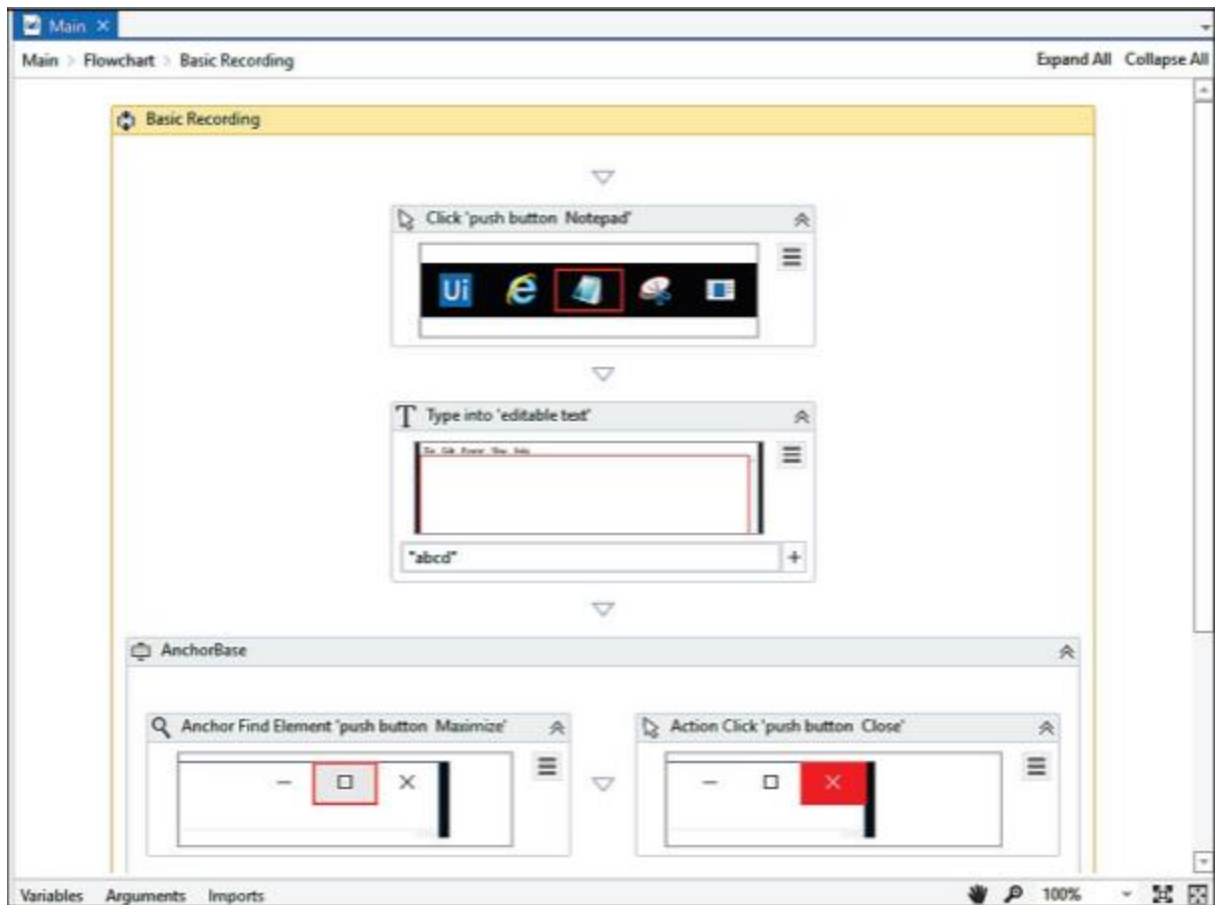
Variable panel

This allows the user to create variables and make changes to them. This is located below the Designer panel. In UiPath Studio, variables are used to store multiple types of data ranging from words, numbers, arrays, dates, times, and timetables. As the name suggests, the value of the variable can be changed. An important point to note is that variables can only be created if there is an activity in the Designer panel. To create new variables, you can go to the DESIGN tab on the Ribbon and click on create variable, then choose the type of variable. Otherwise, one can simply go to the Variable panel located below the Designer panel and create a variable. Also, if one renames a variable in the Variables panel, the variable is renamed in every place it is used in the workflow. The Scope of the variable shows where the variable is located.

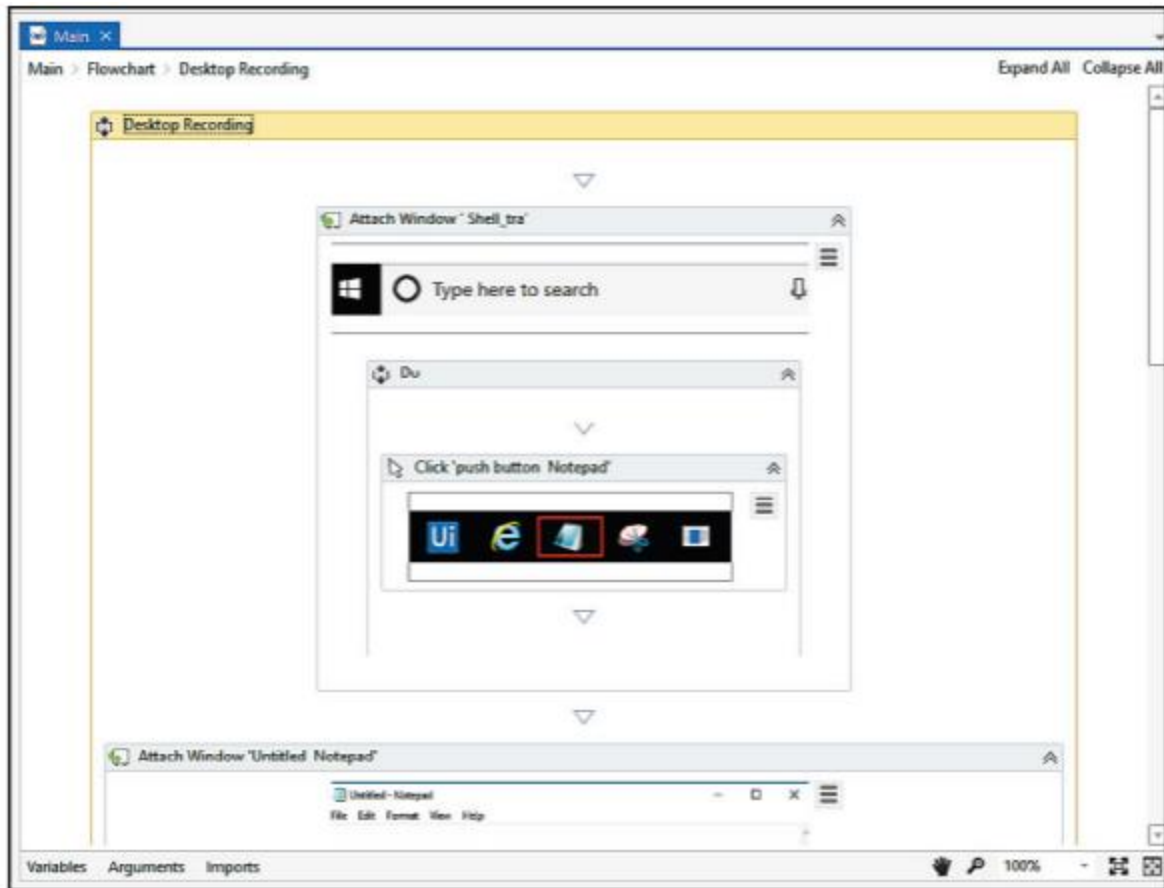
Q.18 What are the different types of recordings available in UiPath Studio? Explain any two.

Ans) There are four basic types of recording:

1. Basic recorder: Basic recorder is used to record activities on the desktop. This type of recorder is used for single activities and simple workflows. The actions here are self-contained and not contained in separate windows, as shown in the following screenshot:



2. Desktop recorder: The desktop recorder, like the basic recorder, is used to record activities on the desktop. However, it is used to record and automate multiple actions and complex workflows. Each activity here is contained in an Attach Window component, as shown in the following screenshot. The Attach Window component is especially important to ensure that other windows of the same application do not interfere in the workflow. UiPath uses the name of the app, the title of the window, and the currently opened file to locate and identify the correct window. However, there may be cases where, for example, two untitled Notepads are open on the screen. Without Attach Window, UiPath may select the wrong Notepad, thus causing errors:



3. Web recorder: The web recorder, as the name suggests, is used to record actions on web applications and browsers.
4. Citrix recorder: Citrix is used to record virtual machines, VNC, and Citrix environments. This recording allows only keyboard, text, and image automation.

Q.19 Compare and contrast manual and Automatic recording.

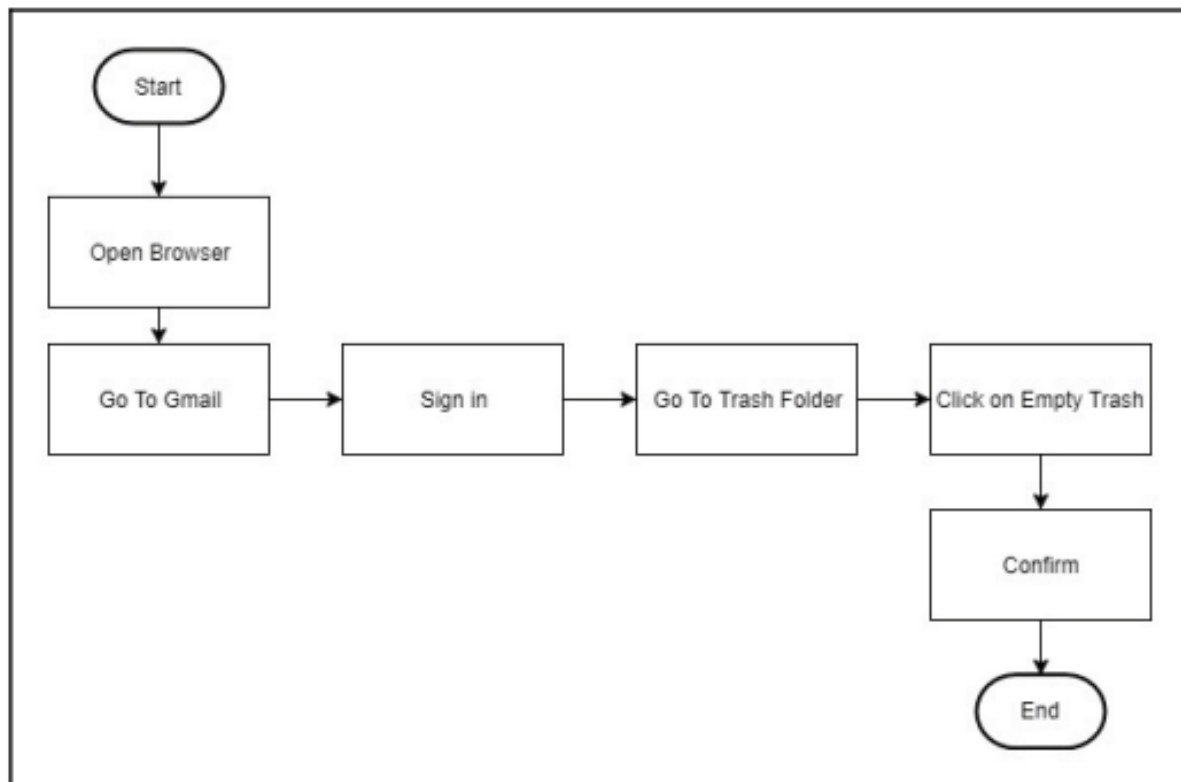
Ans) There are two types of recording:

1. Automatic recording: This is for recording multiple actions in one go. This is a very good feature for preparing a solid foundation for automating a task. It can be invoked with the Record icon available in basic, desktop, and web recorders. The Citrix recorder does not support automatic or multiple step recording. A few types of action cannot be recorded using automatic, for example, hotkeys, right click, double-click, and a few more. For all these activities, you should use a single step recorder, also known as a manual recorder.

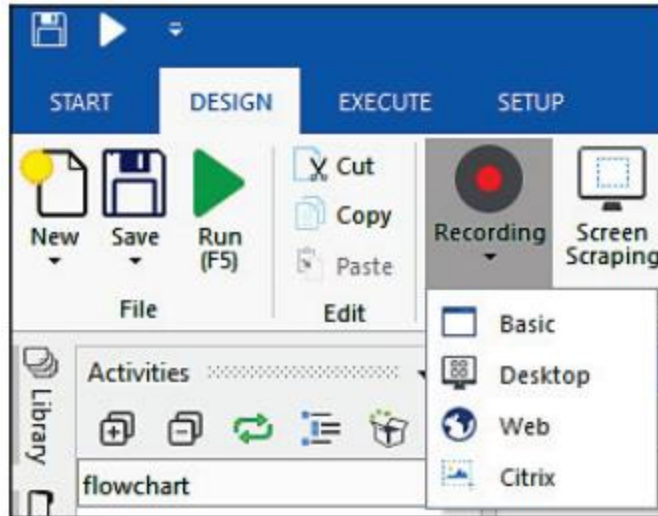
2. Manual recording: This type of recording is used to record each step one at a time and hence offers more control over the recording. Also, it can record all actions that cannot be recorded using automatic recording such as keyboard shortcuts, mouse hover, right-click, modifier keys, such as Ctrl and Alt, finding text from apps, and many other activities.

Q.20 Give the process flow of emptying trash in Gmail.

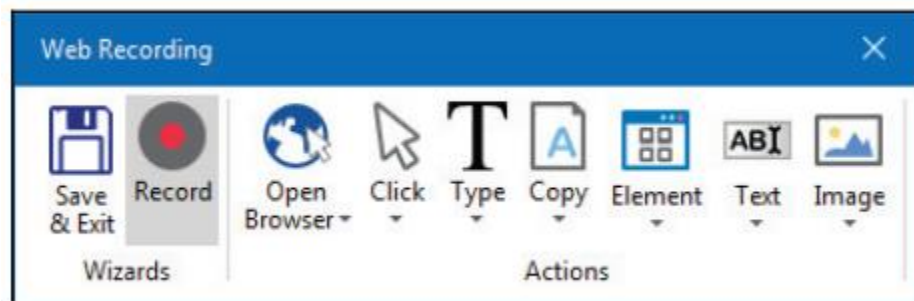
Ans)



First and foremost, we begin with a blank project in UiPath Studio and then choose Web recorder from the Recording drop-down list:

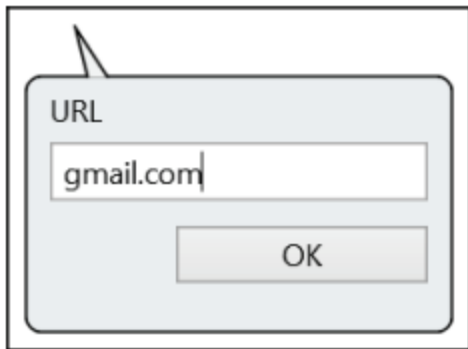


We must click on the Recording option and select the type of recording. As discussed before, we will use Web recording for this process since we are working on a website. Just click on the Recording icon at the top of the page. From the four types of recording that appear, choose Web recording. A Web Recording panel will appear, as shown in the following screenshot:



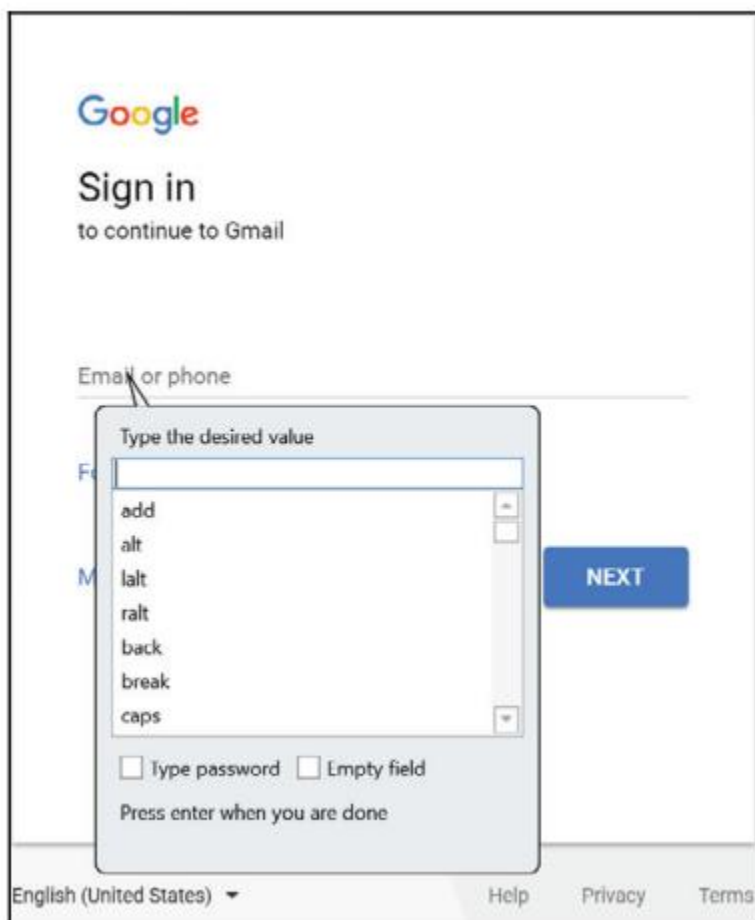
The following are the six steps in our process flow:

1. Open Browser: Although we have already opened Gmail in the browser, we did not record that step. Here, we will note that step in the recorder using the Open Browser button in the recorder. A drop-down menu will appear. Again, choose Open Browser from the drop-down menu. It will ask to highlight the browser, highlight the already opened browser and click on the top of the browser.
2. Go to gmail.com: You will be prompted to enter the URL of the website then navigate to. Type <https://gmail.com> or gmail.com and press OK:



Please remember the first step will merely make note of the steps in the recording but will not do anything on the screen. From the next step onwards, we will use the already opened gmail.com.

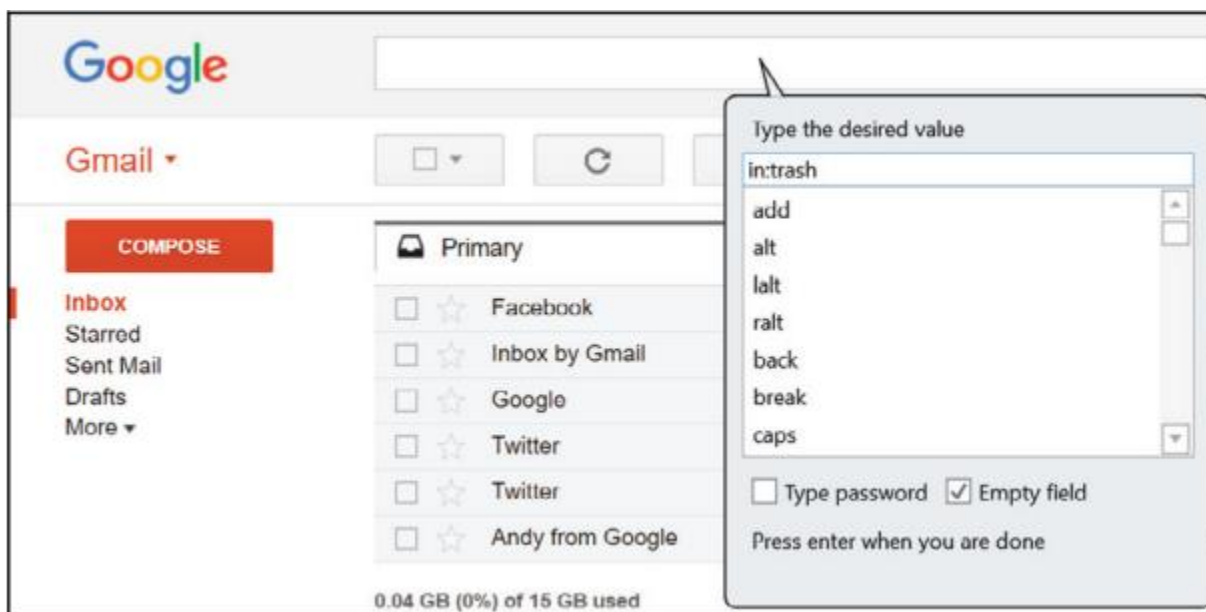
3. Sign In: Start recording by clicking on the Record icon of the recording panel
Go to the already open Gmail and click on the Email or Phone field. UiPath will pop up a prompt for typing the email:



Type email in the box provided by the UiPath recorder and press Enter. The Gmail textbox will automatically fill up with your typed content. Click on the NEXT button of the Gmail interface; it will also get recorded. Now, you have recorded an entry in the password field. For simplicity, you may type the password in the prompt provided by UiPath.

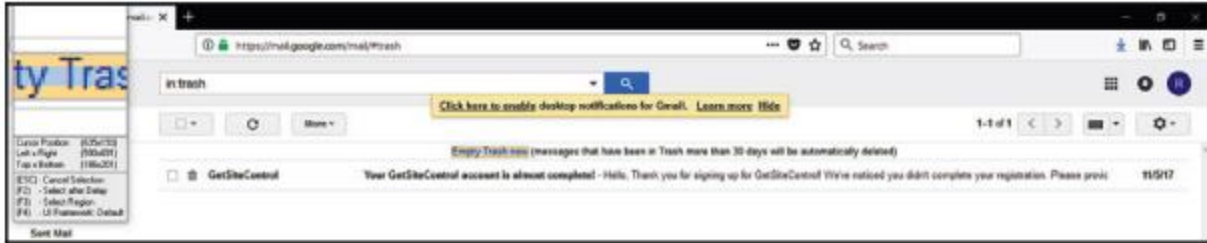
Type your password in the text field of the popup that appears. Then, click NEXT to log in to your account. Clicking on the NEXT button will also get recorded.

4. Locate Trash Folder: In this step, we have to click on the search box of Gmail and type in:trash in the UiPath prompt and hit Enter:



Now, click on the Search button beside the search box. It will also get recorded automatically and the Trash folder will appear.

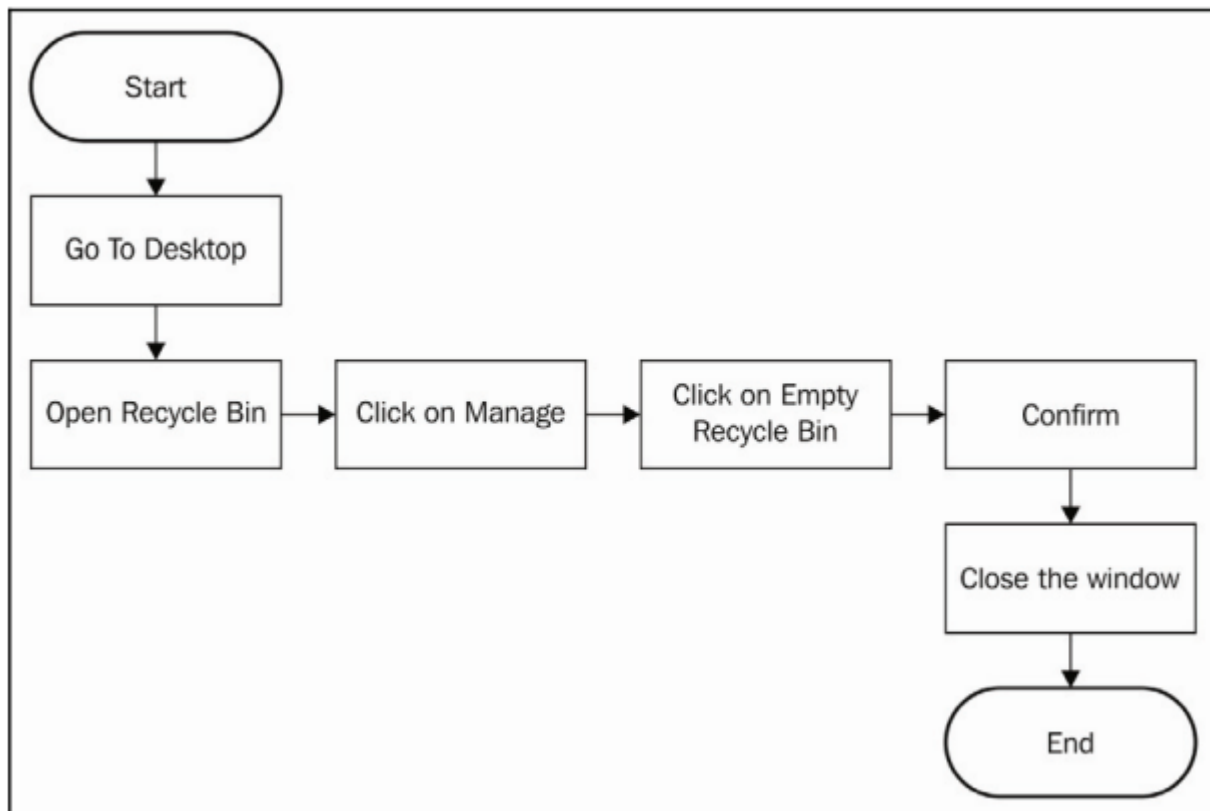
5. Click on Empty Trash now: Once you are done with clicking on the Trash action, You can see a link showing Empty Trash now. Hover mouse on this link and it will get highlighted, click on it to delete all the messages in the Trash folder:



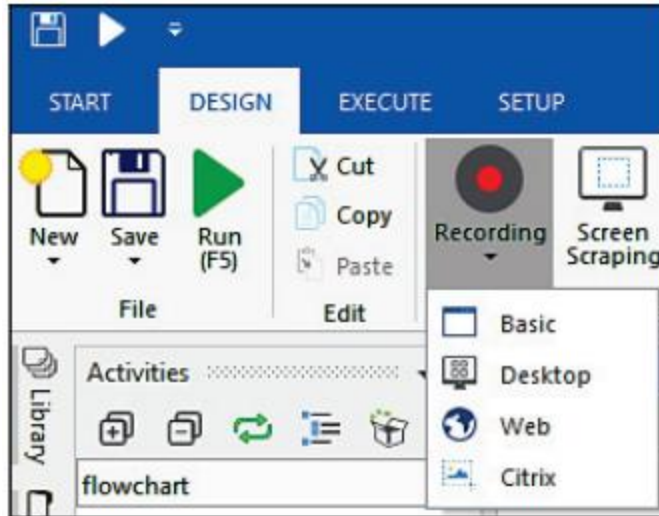
6. Confirm: When you click on Empty Trash now, a confirmation dialog will appear asking your permission for the action. Just confirm your action by clicking on the OK button.

Q.21 Give the process flow of emptying recycle bin.

Ans)



Open UiPath Studio and choose a blank project. Since we are working in the recorder, and since we are working on the desktop and not a web application, we are required to choose the desktop recorder:

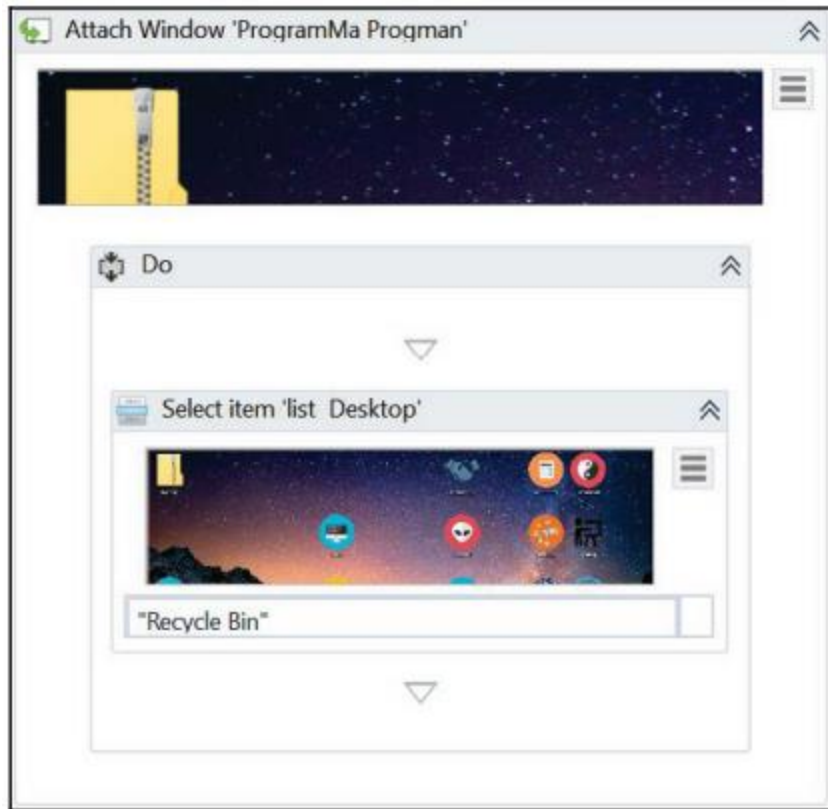


Start the recorder and simply perform the following steps:

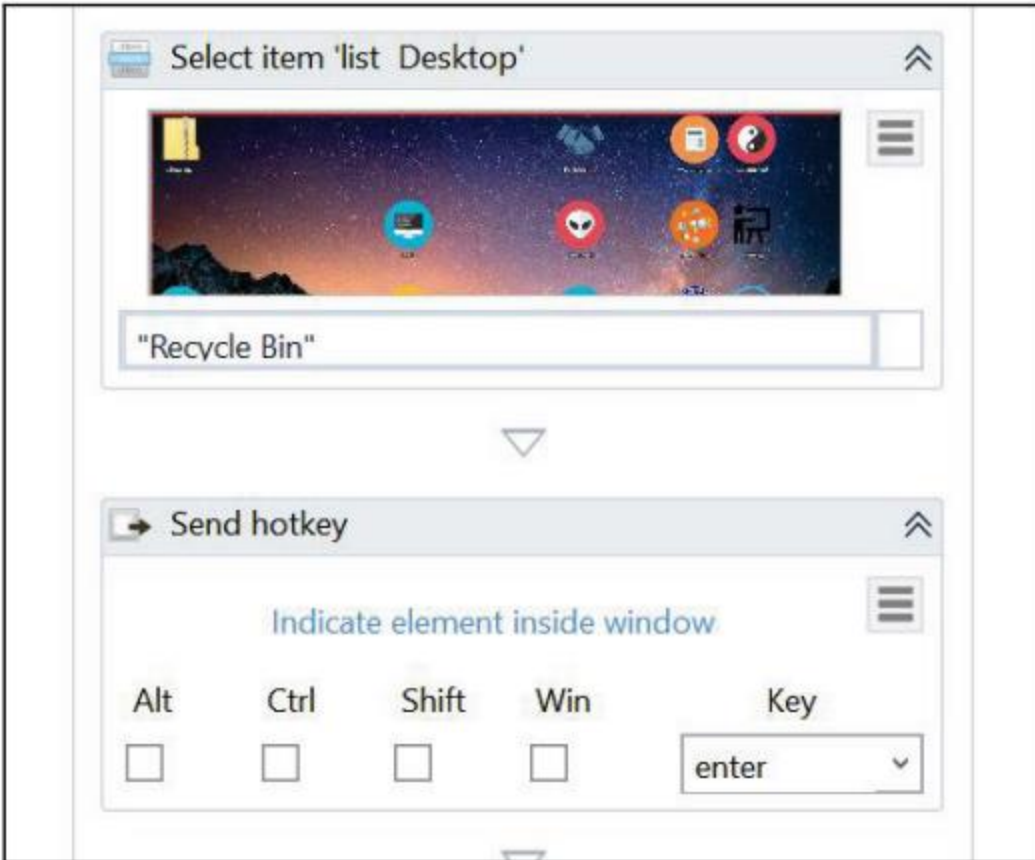
1. Go to the desktop by pressing the Windows + D keys.
2. Open Recycle Bin by clicking on Recycle Bin and then pressing Enter key.
3. Click on the Manage tab of the 3FDZDMF#JO folder.
4. Click on the Empty Recycle Bin button.
5. Confirm by clicking on the Yes button in the dialog box.
6. Close the Recycle Bin folder by pressing the cross button.
7. Press the Esc key and Save & Exit the recorder.

Now your recording is ready to view, let's examine each step recorded:

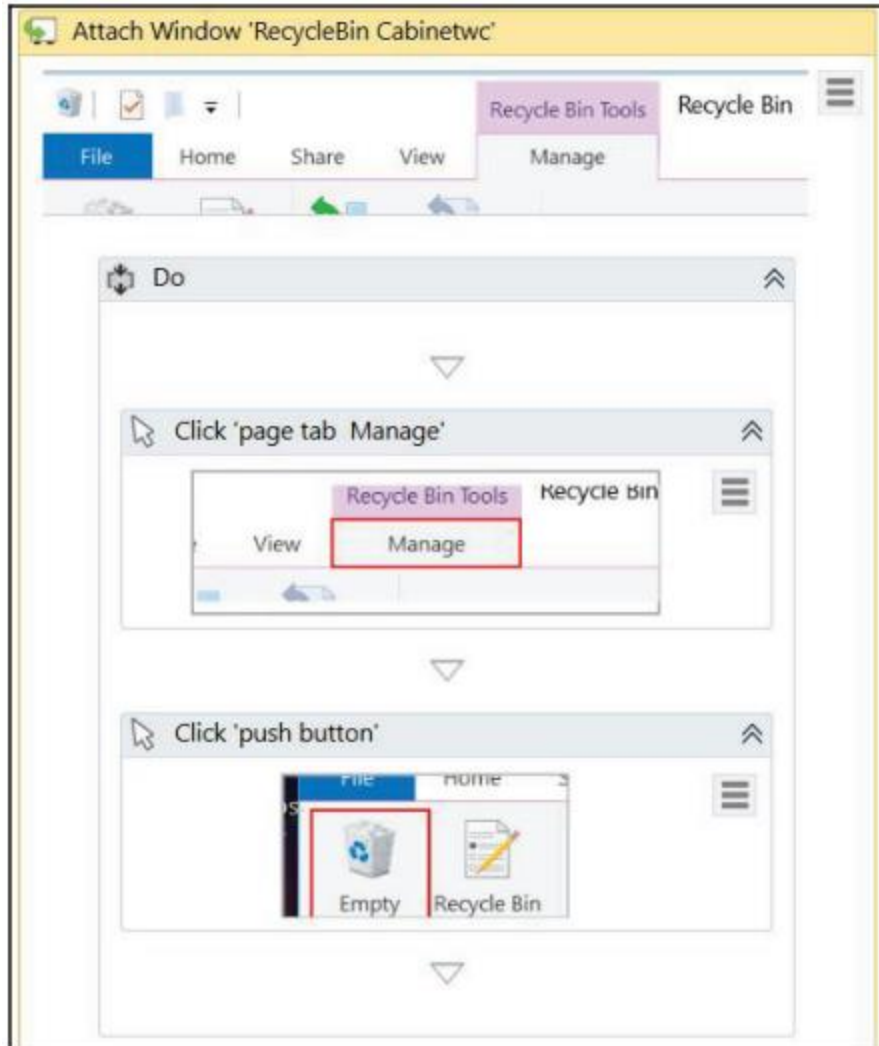
1. Go to the desktop by pressing Windows + D keys: Please note that the recorded steps attach themselves to an application, and execute commands for that application, so the next step (Open Recycle Bin) will be executed on the desktop whether you are there or not.
2. Open Recycle Bin by clicking on Recycle Bin and then pressing the Enter key--We can see the recorded step in the following screenshot:



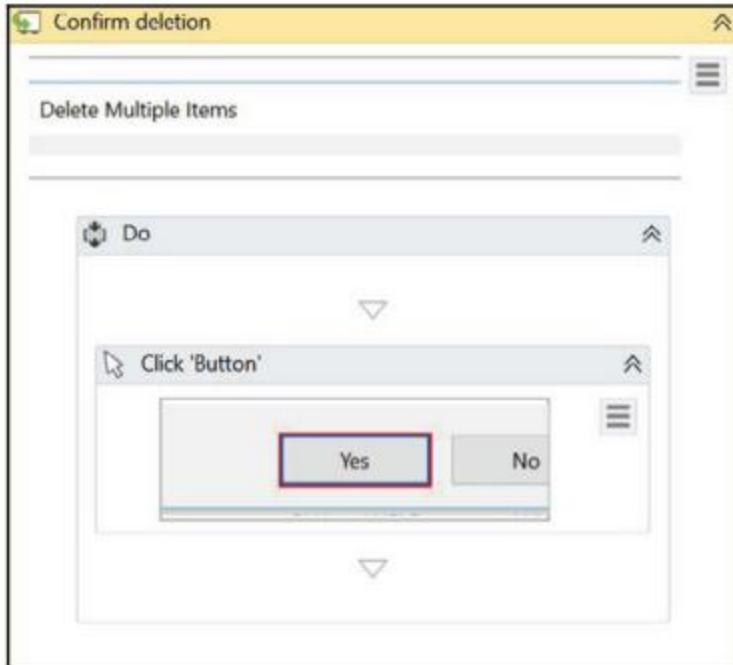
Please note that only selecting the Recycle Bin is recorded, not the Enter key. We should manually add that step. Search for 4FOEIPULFZ in the Activities window and insert it into the workflow just below the Select item 'list Desktop' step, as shown in the following screenshot:



3. Click on the Manage tab of the Recycle Bin folder: This is recorded as it is and so is the fourth step, click on the Empty Recycle Bin button:



4. Confirming by clicking on the Yes button on the dialog box is also recorded smoothly:



In the last step, closing the Recycle Bin folder by pressing the cross button, you may have to indicate an anchor. Save it and press F5 to run it.

Graded Questions

- Q.1 What is automation?
- Q.2 What should be automated?
- Q.3 What can be automated?
- Q.4 What are the techniques of automation?
- Q.5 What is Robotic Process Automation?
- Q.6 What can RPA do?
- Q.7 What are the industries that can benefit from RPA?
- Q.8 What are the benefits of RPA?
- Q.9 What are the components of RPA?
- Q.10 What are the different types of RPA platforms?
- Q.11 Write a short note on UiPath?
- Q.12 What are the components of UiPath? OR Explain UiPath Stack.
- Q.13 What are the different variations available for RPA?
- Q.14 What are the different types of Robots?
- Q.15 What are the main capabilities of Orchestrator?
- Q.16 What are the main types of projects supported by UiPath Studio?
- Q.17 What is user interface in UiPath Studio? What are the different panels

available in UiPath studio? Explain any five.

Q.18 What are the different types of recordings available in UiPath Studio?
Explain any two.

Q.19 Compare and contrast manual and Automatic recording.

Q.20 Give the process flow of emptying trash in Gmail.

Q.21 Give the process flow of emptying recycle bin.